

Enabling Large Language Models to Generate Text with Scientific Paper Citations: QASA Dataset

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Abstract

Retrieval-augmented generation (RAG) systems have become a prominent approach for grounding large language model (LLM) outputs in external knowledge, yet the faithfulness of in-context citations generated by these systems remains underexplored. We adapt and extend the citation evaluation framework of [Gao et al. \(2023\)](#), evaluating two instruction-tuned LLMs, Gemma-4-26B and LLaMA-3.3-70B Instruct, across four prompting strategies (VANILLA, VANILLA + Chain of Thought (CoT), VANILLA + Tree of Thought (ToT), and CLOSEDBOOK) on QASA, a long-form scientific QA dataset requiring deep, grounded reasoning. Using both NLI-based and human evaluation, we find that Gemma consistently outperforms LLaMA, and that multi-step reasoning improves citation quality for Gemma while degrading it for LLaMA, suggesting that its effectiveness is model-dependent. ToT achieves the strongest citation recall and precision for Gemma while maintaining a more balanced recall-precision tradeoff than CoT. We further validate these findings on ELI5, a noisier community-sourced dataset, finding that ToT improvements are still present but less pronounced in open-ended, low-quality retrieval settings. Additionally, we identify limitations of NLI-based evaluation for longer model outputs and propose directions for future work including end-to-end pipeline optimization and further exploration of multi-step reasoning strategies.

1 Introduction

Retrieval-augmented generation (RAG) systems have become a prominent paradigm for grounding large language model (LLM) outputs in external evidence. However, evaluating whether generated responses are both correct and faithfully supported by cited sources remains an open challenge.

In this work, we extend the ALCE citation evaluation framework ([Gao et al., 2023](#)) to the QASA

dataset, a scientific paper QA benchmark requiring long-form, evidence-grounded responses from domain experts. We evaluate two recent open-weight models, Gemma-4-26B and LLaMA-3.3-70B Instruct, across four prompting strategies: VANILLA, VANILLA + Chain-of-Thought (CoT), VANILLA + Tree-of-Thought (ToT), and CLOSEDBOOK with post-hoc citation. We adopt ALCE’s citation quality metrics with targeted modifications suited to our evaluation scale, and replace the original SurgeAI human evaluation with our own manual annotations. Our results show that multi-step reasoning improves citation quality for Gemma but degrades it for LLaMA, and that ToT achieves the strongest overall citation performance on QASA.

2 Related Work

ALCE ([Gao et al., 2023](#)) introduced the first reproducible benchmark for automatically evaluating citation quality in LLM generations, proposing metrics across three dimensions: fluency, correctness, and citation quality. While ALCE demonstrated strong correlations between automatic and human evaluation, it was designed around a specific set of datasets (ASQA, QAMPARI, ELI5) and models available at the time. Subsequent work has explored citation grounding in longer, more complex settings ([Li et al., 2024](#); [Zhang et al., 2024](#)), but these approaches focus on other angles such as chatbot hallucination and long-context Supervised Fine-Tuning (SFT) respectively, leaving scientific long-form QA underexplored. QASA ([Lee et al., 2023](#)) addresses this gap with a benchmark of expert-answered questions over full AI/ML papers, requiring deeper reasoning than factoid or community QA datasets, but does not explore citation capabilities.

3 Proposed Exploration

We investigate the following research questions:

(1) Does the model perform better on the new challenge of scientific paper citation versus the original paper’s focus on the challenges of technical breakdown (ELI5) and ambiguous factoid questions (ASQA)?

We explore a new dataset QASA that tackles a different question of (Lee et al., 2023). Question answering datasets such as ELI5 and ASQA tackle factoid or shallow QA without asking deeper understanding. Conversely, QASA was created to require more sophisticated rationale and reasoning. The QASA benchmark consists of 1798 novel question answering pairs, the questions having been asked by regular readers of AI/ML papers and answered by AI/ML experts. The researchers guided readers and authors to ask questions while reading the whole paper rather than gathering only extractive questions from paper abstracts. For each question, experts wrote detailed, long-form answers, identifying relevant paragraphs as evidence and generating evidential rationales, then finally combining them into a comprehensive final answer. There are an average of around 50 passages from scientific papers provided for each question.

(2) Does modifying the original paper’s prompting strategies improve citation quality and answer correctness?

For our baselines, we reproduce the paper’s VANILLA prompting strategy and CLOSED-BOOK post-hoc citation across two new models. We also add a new strategy where we improve VANILLA through a two-turn process "Chain of Thought." This strategy attempts to increase model awareness, forcing the model to better evaluate its choices and produce a more aware, evidence-grounded answer to the question (more details are provided in the Implementation Details section). Our Chain-of-Thought prompting is structured to force a direct correlation of the reasoning with the citation decisions in the final output.

We add a further extension to VANILLA: Tree-of-Thought, inspired by the single-prompt Tree-of-Thought framework in Dave Hulbert’s paper "Using Tree-of-Thought Prompting to boost ChatGPT’s reasoning." The key idea is for the model to explore multiple branches of reasoning while self-evaluating as it progresses to output a final answer (Hulbert, 2023).

Concretely, we simulate three expert analysts: a

Document Analyst, a Fact Checker, and an Answer Synthesizer. Each of them independently assesses document relevance and propose claims across two rounds of reasoning before converging on a final cited answer, encouraging the model to explore and reconcile multiple reasoning paths rather than committing to a single chain of thought.

(3) Does updating the original paper’s models to newer models improve citation performance?

In the original paper, they used the closed-weight models gpt-3.5-turbo-0301, gpt3.5-turbo-16k-0613, and gpt-4-0613, For open-source models, the researchers tested LLaMA and its instruction-tuned versions, including Alpaca, Vicuna, Oasst. We updated these models to more recent versions and decreased to 2 models because of time constraints. The two models we used are Gemma-4-26B (google/gemma-4-26b-a4b-it) and LLaMA-3.3 70B (meta-llama/llama-3.3-70b-instruct). Both models were instruction-tuned and evaluated using an identical set of 200 prompts (we cut down to 200 because of time and credit constraints). Each model is tested across four prompting strategies: Vanilla, Vanilla with Chain-of-Thought (CoT), Vanilla with Tree-of-Thought (ToT), and closed-book with post-hoc citation.

We benchmark these two competitive open-weight models under identical conditions, providing a direct comparison of their ability to retrieve-and-cite versus generating text with citations.

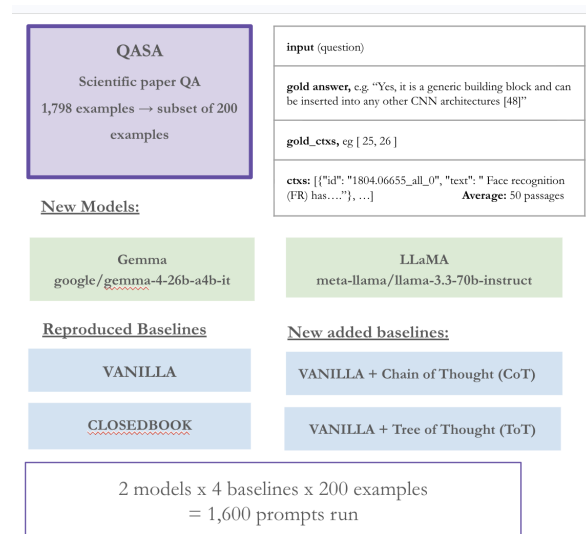


Figure 1: Proposed Exploration

4 Implementation Details

4.1 Prompt Construction

Vanilla and Vanilla with CoT were limited to using one-shot demos in their prompting strategies due to OpenRouter’s token limits for the input to prevent empty model outputs. Closed-book, on the other hand, used three-shot demos.

4.1.1 CLOSEDBOOK

We replicate the CLOSEDBOOK baseline from the original paper. The instruction was simplified to *"Write an accurate and concise answer for the given question,"* with no passages provided as context ($\text{ndoc} = 0$). Post-hoc citations were then assigned after generation using the same GTR-based retrieval used in the Vanilla setting.

One notable implementation difference arose around the system message. Despite removing citation requirements from the prompt instruction, we found it necessary to update the system message from *"You are a helpful assistant that answers the following questions with proper citations."* to *"You are a helpful assistant that answers the following questions concisely and accurately."* Without this change, the two models behaved differently: Gemma-4-26B correctly followed the prompt instruction and produced citation-free answers even with the original system message, while LLaMA-3.3 70B continued to generate nonsensical bracket citations despite receiving no documents ($\text{ndoc}=0$). This suggests LLaMA adheres more strictly to system-level instructions than to the prompt.

4.1.2 VANILLA prompting

We retrieve the top 3 passages per question using GTR similarity (sentence-transformers/gtr-t5-large), decreased from the original paper’s top 5 to accommodate QASA’s longer passage lengths within prompt token limits. We exclude examples with more than 3 gold passages, as the top-3 retrieval budget cannot guarantee all gold passages are present. Passages are relabeled [1][2][3] in display order, and gold answer citations are remapped to match this new indexing. Any gold passage outside the top 3 is swapped in by evicting the lowest-ranked non-gold passage to isolate citation capability from retrieval capability, though we note this may inflate recall relative to a true end-to-end pipeline.

"Write an accurate and concise answer for the given question using only the provided search results. Always cite factual claims with bracket citations such as [1][2]. Use at least one citation per sentence and at most three citations per sentence."

4.1.3 VANILLA with Chain of Thought Citation Awareness

Our chain-of-thought (CoT) prompting strategy is an extension of VANILLA, providing the top 3 passages retrieved by GTR and extending it with two additional rounds. In the first round, the model is presented with the question and retrieved passages and asked to explicitly reason about each passage before generating an answer: the model is prompted to assign a relevance level (high, medium, or low) and a justification, then identify the 1–3 most critical passages by rank. This reasoning trace is then fed back into a second prompt as a "Prior analysis" block, conditioning the final answer generation on the model’s own deliberation. The final answer is required to cite passages by rank (e.g. [1][2]), with at least one and at most three citations per sentence with only a minimal sufficient subset, which aligns with the original paper’s precision metric goal of penalizing irrelevant over-citations. The full instruction is in Appendix A.2.

4.1.4 VANILLA + Tree-of-Thought:

We use the same setup as VANILLA (GTR retrieval, gold swapping, remapping, etc.). However, we change the instruction and the expected output format to incorporate Tree-of-Thought logic. Rather than prompting the model to produce a direct answer, we prompt it to simulate three expert analysts: a Document Analyst (Expert A), a Fact Checker (Expert B), and an Answer Synthesizer (Expert C). These experts collaborate across two structured rounds of reasoning before converging on a final cited answer. In Round 1, each expert independently assesses the relevance of each retrieved document. In Round 2, each expert proposes, fact-checks, and synthesizes key claims supported by the documents. The final Answer is then produced as a single, fluent paragraph with bracket citations. Our instruction to the model is in Appendix A.3.

4.2 Evaluation

Subclaim Extraction. To evaluate answer correctness, we extract 1–3 subclaims from each

gold answer using GPT-5.4 mini via OpenRouter. We use few-shot prompting with 4 manually constructed demonstrations, with 4 selected empirically based on how many could fit within the prompt context while covering a range of questions that had gold answers with varying numbers of citations. Each demonstration consists of a question, a gold answer, and manually written subclaims. The model is then prompted with the instruction: *"Read the question and the gold answer. Write 1–3 concise subclaims directly supported by the gold answer only. Use lines starting with Subclaim 1:, Subclaim 2:, ..."* The extracted subclaims are then used to compute answer correctness via NLI entailment, following a recall and precision evaluation similar to the original paper, described below.

NLI model. For a more lightweight model to run on the A100 GPU on Colab, we use the facebook/bart-large-mnli model from HuggingFace (<https://huggingface.co/facebook/bart-large-mnli>), loaded via the transformers pipeline for text classification.

Citation Recall and Precision. The model takes as input a concatenation of the premise and hypothesis in the format {PREMISE} {HYPOTHESIS}" and outputs an entailment label with a confidence score. We format each passage used as a premise as TITLE\nTEXT" and concatenate all passages with a newline separator. Our implementation follows the original ALCE framework in using hard binary scores: the NLI model returns 1 if the entailment label is predicted with confidence above a threshold of 0.5, and 0 otherwise. These scores are averaged across statements and citations to compute recall and precision respectively. The overall structure of the metrics remains identical to the original paper: recall checks whether the joint set of cited passages entails the statement, and precision checks whether each individual citation is non-redundant, with precision set to 0 whenever recall is 0.

Claim Recall. We replicated the Claim Recall evaluation performed on ELI5 in the original paper for QASA. For each question, claim recall is computed by running NLI with the model’s output (stripped of citations) as the premise and each gold extracted subclaim as the hypothesis, using the facebook/bart-large-mnli model with a hard binary threshold of 0.5 described above. A subclaim receives a score of 1 if the model predicts entailment with confidence above 0.5, and 0 other-

wise. Claim recall for each item is then the average of these binary scores across all its subclaims. This measures whether the model’s answer covers the key facts in the gold answer, as opposed to citation recall and precision which measure whether the cited passages support the model’s statements.

Since QASA supports long text responses varying in length and amount of sub-claims made, we excluded the EM Recall criteria from our evaluations as they score citation quality from the frequency of exact-sub string matches between the gold output and the model-generated output. Since QASA isn’t shallow QA/factoid-based, this metric is irrelevant for our study.

MAUVE. We ran MAUVE similar to the paper to evaluate output fluency, but we noted that this was likely not going to be a productive metric due to the length difference between the gold answer and the model output (model output tended to be more verbose).

Manual Analysis For manual evaluation, we define precision and recall to mirror the automated NLI metrics. Recall was computed as the number of citations that entailed their respective claims divided by the total citations in the model output, counting each citation occurrence individually. Precision penalizes irrelevant and redundant citations: we first assessed each statement for relevance, then evaluated whether its citations were justified. We followed the original paper’s stipulation that precision is 0 if recall is 0, and precision can only be 1 if recall is 1. We also performed Utility scoring the same as in the original paper (answering if "The response is a helpful and informative answer to the query" on a Likert scale of 1-5, corresponding to Strongly Disagree, Disagree, Neutral, Agree, and Strongly Agree).

We manually evaluated the first 20 questions of the dataset for all settings. Spreadsheets with scoring justifications are linked in Appendix A.1.

5 Results

We present the citation quality scores via NLI on the QASA dataset for all four prompting strategies in Table 1. We present the citation quality scores via manual human-evaluation on the QASA dataset for all four prompting strategies in Table 2. Note that for ToT, only Gemma was evaluated. This is because with NLI and manual evaluation, we found that the CoT extension degraded LLaMA

citation quality (precision and recall) significantly (more on this in the Discussion section), so we decided to only attempt the even more involved ToT framework on Gemma.

Findings are analyzed in-depth in the Discussion section.

Table 1: NLI Evaluation on QASA Dataset

Strategy	Model	Recall (%)	Precision (%)	Claim Recall (%)
Vanilla	Gemma	75.25	72.68	40.75
	LLaMA	62.30	58.49	42.00
CoT	Gemma	77.31	75.62	50.50
	LLaMA	47.48	45.18	51.08
ToT	Gemma	78.73	77.28	35.92

Table 2: Human Evaluation on QASA Dataset

Strategy	Model	Recall (%)	Precision (%)	Utility
Vanilla	Gemma	92.85	91.35	4.25
	LLaMA	87.70	81.00	3.85
CoT	Gemma	99.13	65.49	4.65
	LLaMA	93.34	50.45	4.1
Closed Book	Gemma	N/A	N/A	2.7
	LLaMA	N/A	N/A	2.6
ToT	Gemma	97.50	86.35	4.35

5.1 MAUVE

Model	Setting	MAUVE
LLaMA-3.3-70B	Closedbook	78.65
LLaMA-3.3-70B	Vanilla	36.89
LLaMA-3.3-70B	Vanilla w/CoT	4.49
Gemma-4-26B	Closedbook	84.48
Gemma-4-26B	Vanilla	43.12
Gemma-4-26B	Vanilla w/CoT	34.85

Table 3: MAUVE scores (300-word truncation) across models and generation settings on QASA (200 examples). Higher is better.

Table 3 reports MAUVE scores measuring distributional similarity between Gemma and LLaMA model outputs and the gold reference answers. MAUVE scores were highest for CLOSEDBOOK outputs (Gemma: 84.48, LLaMA: 78.65), likely because their concise style most closely matches the short gold references rather than reflecting superior factual grounding. CoT outputs scored lowest (Gemma CoT: 34.85, LLaMA CoT: 4.49), as CoT responses averaged over 100 words against gold references averaging 15 words. Given MAUVE’s

sensitivity to output length, we treat it as unreliable for our citation-focused evaluation and do not report it as a primary metric.

6 Discussion

6.1 QASA Dataset

6.1.1 VANILLA Analysis

Table 1 reports citation recall and precision for both models in the VANILLA setting. Gemma-4-26B outperforms LLaMA-3.3-70B on both metrics, achieving 75.25% recall and 72.68% precision compared to LLaMA’s 62.30% and 58.49% respectively. This suggests that Gemma is both more likely to support its claims with relevant citations and less likely to include unnecessary ones, while LLaMA tends to overcite.

Qualitative inspection of high-scoring examples supports this finding. A clear example illustrating Gemma’s higher precision is the question “Is hyperparameter optimization performed independently for the two dataset corpora?” Gemma responded concisely with “Yes, all model hyperparameters were tuned on the respective validation sets of the two corpora [1],” earning perfect recall and precision with a single citation that entailed the statement and a short, concise answer.

Conversely, LLaMA gave the correct answer in its first sentence but continued to give unnecessary detail: “Hyperparameter optimization is performed independently for the two dataset corpora...[1]. The hyperparameters considered for tuning include hidden layer sizes, depths, initial learning rates, batch sizes, and dropout...[1]. The experimental setup and results are provided in more detail in Appendix A and Table 5, respectively [1].” LLaMA’s output repeatedly cites [1]; while citations were not strictly incorrect (e.g. the cited passages do still entail the statements), the same document appearing three times in adjacent sentences for closely related sub-facts is clear evidence of over-citation. The additional sentences add little to the answer while inflating the citation count and leading to a more verbose answer that is more difficult for the reader to understand. This example illustrates a broader pattern where LLaMA’s tendency toward verbose responses hurts its precision score, while Gemma’s more concise outputs score higher by citing only what is necessary to support each claim without sacrificing answer relevance and completeness (see full output without ellipses in A.4).

Case Study. We perform an interesting case study on a question where the retrieved documents simply do not contain the answer. For “What is the full form of ILSVRC?”, the gold answer was that “since there is no information about the form of ILSVRC in this paper, this question cannot be answered and requires external knowledge.” However, rather than abstaining, both models hedged while still attaching citations, earning zero recall and precision on the NLI evaluation. LLaMA answered that “the full form of ILSVRC is not explicitly stated in the provided documents [1][2][3]; however, based on the context, it can be inferred that ILSVRC refers to a competition or dataset related to image classification,” speculatively extending beyond what the documents support. Gemma was more restrained, stating “based on the provided documents, the full form of ILSVRC is not mentioned,” but still cited all three documents despite none of them supporting the claim (see Vanilla Example 2 in A.4 for full output). This suggests a shared limitation: neither model reliably detects when the retrieved context is insufficient, defaulting instead to citing loosely related passages and offering speculative interpretations rather than cleanly acknowledging the gap. This behavior is indicative of a sycophantic tendency: an inability to refrain from attempting to satisfy the asker even when the honest answer is that the question cannot be answered from the available evidence.

Output Failures. Moreover, NLI-based citation evaluation also had its own limitations. In the hyperparameter question discussed above, Gemma’s answer “all model hyperparameters were tuned on the respective validation sets of the two corpora [1]” was a faithful paraphrase supported by the cited document. However, NLI scored it as zero on both recall and precision. The NLI pipeline failed to recognize that this phrasing entails the gold answer’s meaning, suggesting that NLI-based metrics may underestimate citation quality when models paraphrase rather than reproduce source text verbatim.

Manual Analysis Our manual analysis of the first 20 prompts for both models returned results consistent with the NLI scores: LLaMA achieved lower recall, precision, and utility scores than Gemma, with a 5 percentage point difference in recall, 10 percentage point difference in precision, and a 0.4 difference in utility. The primary driver of this gap was verbosity (as discussed above), as

LLaMA’s answers tended to be substantially longer and less focused.

6.1.2 VANILLA with Chain of Thought Citation Awareness Analysis

Table 1 reports citation recall and precision for both models in the VANILLA + CoT setting under NLI evaluation. Gemma-4-26B significantly outperforms LLaMA-3.3-70B on both metrics, achieving 77.31% recall and 75.62% precision compared to LLaMA’s 47.48% and 45.18% respectively.

Compared to VANILLA, citation recall and precision scores improved from 75.25 to 77.31 and 72.68 to 75.62, respectively for Gemma-4-26B. This finding provides evidence that chain-of-thought prompting on full scientific texts can yield higher citation quality by forcing the model to engage in multiple steps of reasoning that may help prevent the model from hallucinating claims from the passages and align citation contexts to their extracted claims, grounding the final output.

However, this hypothesis falls apart when considering the citation recall and precision scores for the LLaMA 3.3-70B Instruct model, dropping significantly from 62.30% to 47.48% and 58.49% to 45.18%, respectively, when switching from VANILLA prompting to VANILLA + CoT.

Human Evaluation. Table 2 reports citation recall and precision for both models in the VANILLA + CoT setting under human-evaluation. Qualitative inspection affirms that under VANILLA + CoT, Gemma-4-26B tends to outperform LLaMA 3.3-70B Instruct on three metrics: recall, precision, and utility. Specifically, Gemma-4-26B achieved a near-perfect recall score of 99.13%, a precision score of 65.49%, and a nearly perfect utility of 4.65. LLaMA 3.3-70B Instruct achieved a recall score of 93.34 %, a precision score of 50.45% and a utility of 4.1, lagging behind in all metrics but particularly precision. Its lower recall and precision scores can be explained by its tendency to attribute statement to multiple redundant passages rather than isolating the most prominent passage and its failure to find information relevant to answering the question.

To illustrate both cases, consider the following example with question, "In language models, which method would be better for preventing overfitting from batch normalization and dropout" and gold answer, "According to this work, without dropout, a vanilla LM can run the risk of overfitting, which decreases performance" with LLaMA’s full model

output in A.5. Immediately, the model concludes that no information explicitly answering the question can be found in any of the three passages and yet cites all of them, lowering the recall, precision, and utility score by citing irrelevant passages for an erroneous statement. LLaMA-3.3-70B Instruct’s tendency to redundantly summarize its response at the end of the output is also demonstrated, over-bloating the response with irrelevant citations. Compared to Gemma-4-26B, LLaMA-3.3-70B Instruct tends to be more verbose in generating output and lenient in citing multiple passages rather than identifying the minimal subset of passages necessary to support a claim. Stylistically, LLaMA-3.3-70B also prefers to extend the prompting instructions to follow its own preferred formatting style, a behavior that can possibly be attributed to its instruction-tuning during training.

Compared to Gemma-4-26B’s output for the same question, also shown in A.5, the model is comparably verbose, but reduces unnecessary citations, entailing all claims with a perfect recall. However, only the first citation is relevant to answering the question, decreasing precision. In addition, the model maintains the improper LaTeX formatting from the original input in its output.

An example where both models performed well with perfect recall and utility is for question "What point is different in GATs in terms of assigning weight compared to GCN?" in A.5. Both models achieved perfect recall and utility here, though they both lost precision points due to redundant or irrelevant citations, repeating a pattern consistent with the verbosity issues described above.

6.1.3 Tree-of-Thought (ToT) Analysis

We evaluated the Tree-of-Thought (ToT) prompting strategy exclusively on Gemma, as LLaMA’s drastic drop in citation recall and precision under CoT suggested that multi-step reasoning may complicate rather than clarify its outputs, making it a poor candidate for the more demanding ToT framework. Under NLI evaluation, ToT achieved the highest recall and precision of any prompting strategy for Gemma, at 78.73% and 77.28%, outperforming both VANILLA and VANILLA w/CoT. Its claim recall of 35.92%, however, was notably lower than both VANILLA and CoT.

Rather than indicating a weakness in citation grounding, this likely reflects both ToT’s tendency to elaborate beyond the gold answers and a broader limitation of NLI-based evaluation. NLI models

such as facebook/bart-large-mnli are trained to detect entailment between short, self-contained sentence pairs, and tend to rely on lexical overlap and shallow inference. When the premise is a long, multi-sentence output and the hypothesis is a high-level paraphrase supported implicitly across several sentences, the NLI model may fail to detect entailment even when a human reader would find it obvious. For example, for the question “What are the motivations behind choosing TREC-COVID for analysis on annotation bias?”, ToT’s final answer explicitly states that manually annotating missing relevance judgments resulted in significant performance improvements for non-lexical approaches”. This directly supports the gold subclaim that “TREC-COVID was chosen because the dataset could be made unbiased by manual annotations” (see full example "ToT QASA Example" in A.8). Yet, here, the NLI model returned non-entailment, likely because the connection requires abstracting across multiple sentences rather than matching a single contiguous span. This suggests that NLI-based claim recall may systematically underestimate the quality of longer, more elaborated outputs like those produced by ToT.

ToT’s superior performance is supported by the human evaluation results. ToT achieved strong recall and precision (97.50% and 86.35%), and maintained a far more balanced recall-precision tradeoff than CoT, whose precision collapsed to 65.49% despite near-perfect recall of 99.13%. Utility scores were comparable across strategies; given our sample of 20 examples, small numerical differences should not be over-interpreted. Taken together, these results suggest that ToT is a promising direction for further exploration.

6.1.4 CLOSEDBOOK Analysis

Table 2 reports citation recall and precision for both models in the CLOSEDBOOK setting under human-evaluation. CLOSEDBOOK outputs achieved relatively low utility scores (2.7 for Gemma and 2.6 for LLaMA) and were typically unhelpful for factoid questions: both models hallucinated (when asked about an unanswerable question "Which datasets are used by the paper?", Gemma answered "CelebA and LSUN" and LLaMA answered "CIFAR-10 and ImageNet," when the correct answer was UCF-101 and HMDB-51). For theoretical questions requiring general scientific definitions, both models performed better, as their parametric knowledge covers general scientific con-

cepts not tied to a specific paper’s results.

6.2 ELI5 Dataset

We also reran the VANILLA, VANILLA w/CoT, and VANILLA w/ToT baselines on the ELI5 dataset (with ToT being for Gemma only), to see if our new added extensions of CoT and ToT can still improve upon the VANILLA baseline similar to how it did for QASA. Since CoT degraded LLaMA performance sharply, we expected that ToT would likely result in even more overcited/verbose output, so we omitted LLaMA from evaluation here.

Table 4: NLI Evaluation on ELI5 Dataset, Gemma-only

Strategy	Recall (%)	Precision (%)	Claim Recall (%)
Vanilla	60.98	56.98	8.17
CoT	58.36	55.33	9.50
ToT	62.07	58.02	18.83

Table 5: Human Evaluation on ELI5 Dataset

Strategy	Model	Recall (%)	Precision (%)	Utility
Vanilla	Gemma	81.00	46.50	2.7
CoT	Gemma	97.50	42.00	2.7
ToT	Gemma	96.70	38.75	2.3

VANILLA and VANILLA w/CoT analysis. Recall and precision scores for Gemma-4-26B were consistently lower on ELI5 than on QASA across all prompting strategies except CLOSEDBOOK (Table 4). This pattern reflects several dataset-specific factors. First, ELI5 uses a two-step retrieval pipeline (retrieving the top 100 passages, then re-ranking down to 3) compared to QASA’s direct selection of 3 from 50, making it harder to surface relevant context. Second, ELI5’s gold answers are inconsistently formatted and frequently inaccurate: with 270k threads sourced from a Reddit forum ("Explain Like I’m Five") (Fan et al., 2019). The dataset is prone to typos, colloquial phrasing, and unserious responses (see example in A.7).

On QASA, CoT marginally improved citation quality over the VANILLA baseline. On ELI5, however, the trend reversed: VANILLA outperformed CoT on both recall (60.98% vs. 58.36%) and precision (56.98% vs. 55.33%). We attribute this to the open-ended nature of ELI5 questions, which are less grounded than those in QASA. An intermediate reasoning step that explicitly evaluates document relevance may be less useful, or even

counterproductive, when questions admit a wide range of valid answers and retrieved documents are only loosely constrained to the topic. However, there is still evidence that VANILLA + CoT may help shorten a response to only include relevant citations, such as in Example A.7.

ToT analysis. On the NLI evaluation, ToT achieves the highest citation recall (62.07%) and precision (58.02%), and notably the highest claim recall by a large margin (18.83% vs. 9.50% for CoT and 8.17% for VANILLA) (Table 4). This suggests that the structured multi-step reasoning process encourages the model to produce outputs that better cover key facts in the gold answers.

However, human evaluation of ToT precision was low because in the 20 examples from ELI5, many did not have sufficient information to answer the question. In these cases, Gemma still attempted to construct answers by citing tangential details from multiple passages, which introduced many irrelevant statements and reduced overall precision (see "ToT ELI5 Example" in A.8). As a result, these responses were frequently judged as unhelpful (utility of 1). This behavior suggests that the expert-debate structure may encourage over-speculation, forcing an answer when the available evidence is insufficient instead of explicitly acknowledging the lack of adequate information.

Conclusion

Gemma and LLaMA, under the VANILLA strategy, are typically sufficient in answering theoretical questions on full scientific texts requiring long responses, according to their citation quality metrics. There is mixed evidence for the VANILLA + CoT prompting strategy, with Gemma providing evidence that intermediate reasoning steps may be helpful in reducing irrelevant citations and faithfully aligning statements to their cited sources, while LLaMA performs significantly worse on all metrics. CLOSEDBOOK proved unhelpful for answering factoid and shallow QA questions, instead faring better on theoretical questions, though with the limitation that answers were characterized by general, vague scientific definitions.

ToT emerged as the most promising prompting strategy for Gemma on QASA, achieving the strongest NLI-based citation recall and precision while maintaining a more balanced recall-precision tradeoff than CoT under human evaluation. Its lower claim recall under NLI evaluation is likely

an artifact of the metric rather than a true weakness, as NLI models tend to underestimate entailment in longer, elaborated outputs. On ELI5, however, performance across all strategies degraded relative to QASA, reflecting the dataset’s noisier retrieval pipeline and lower gold answer quality, suggesting that the gains from more sophisticated prompting strategies may be harder to realize on open-ended, community-sourced datasets.

For future work, we would like to see further research on the VANILLA + CoT prompting strategy. Promising extensions include classifying each passage as IRRELEVANT, NEUTRAL, or RELEVANT; listing the minimal factual claims needed to fully answer the question; or exploring inter-passage relationships for more holistic reasoning. We would also like to see more state-of-the-art models evaluated under VANILLA + CoT to determine whether LLaMA’s poor performance is the exception or the norm. ToT likewise warrants further investigation, particularly with respect to varying the number of reasoning rounds, the number of experts, and the aggregation strategy, as well as evaluating on more datasets beyond QASA.

Lastly, since we focused on the citation extraction stage of the pipeline, we propose future research into a full end-to-end citation pipeline that jointly optimizes relevant passage extraction and faithful in-text citation generation, and explores larger top-k retrieval values.

Limitations

The QASA dataset from Huggingface sometimes had minor formatting and grammatical issues in the questions, which may have affected the quality of the models’ outputted responses. In addition, some sub-claims from the gold answers do not explicitly correspond to the cited gold passages, worsening NLI-evaluation scores.

Because of time and computational constraints, we had to limit to evaluation of 200 questions per model+strategy combination, instead of the 1k questions per dataset evaluation performed in the original paper. Accordingly, we only manually inspected 20 question-answer pairs per dataset and prompting strategy instead of 40, as proposed in the original study, since we do not have resources to employ workers from SurgeAI and could only perform evaluations ourselves manually.

7 Data Availability

Our code is available at https://github.com/kaylaxu15/nlp_citations.

Ethics Statement

This work builds on the publicly available ALCE repository and extends it to new models and datasets. All datasets used (QASA and ELI5) are publicly available benchmarks on Huggingface. We cite all prior work on which our methods and evaluation frameworks are based. Our manual evaluation was conducted by the authors themselves rather than crowd workers, avoiding concerns around fair compensation for annotators. We do not anticipate broader harms from this work, as it focuses on improving citation faithfulness in LLM outputs, which supports research integrity.

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A Appendix

A.1 Scoring Justifications

Full scoring justifications for all 200 evaluated outputs are provided in the spreadsheets below.

- QASA, Vanilla analysis for Gemma: https://docs.google.com/spreadsheets/d/1LIY7WcbFr-slZLJ1loLFkH9ocq4UxH_pWBfP_N6yzWk/edit?usp=sharing
- QASA, Vanilla analysis for LLaMA: https://docs.google.com/spreadsheets/d/18me2_ETDttJ4QnwOPgvMTH7_2a8TJz7UNAT04HkSJJU/edit?usp=sharing
- QASA, CoT analysis for Gemma: https://docs.google.com/spreadsheets/d/1TLDFMMHTqkMQq0RiDLqmsz3aT-Z2SBpuCULdD_Wzeoo/edit?usp=sharing
- QASA, CoT analysis for LLaMA: https://docs.google.com/spreadsheets/d/1Syfp-mWs7p9081lzCXTfbWLDOfY0Zu_9a0GclCGkv8g/edit?usp=sharing
- QASA, ToT analysis for Gemma: <https://docs.google.com/spreadsheets/d/1Vvm9yHuyX00i3TUXyWozRdeHtli5mf9P/edit?usp=sharing&ouid=104543084022695088057&rtpof=true&sd=true>
- QASA, CLOSEDBOOK analysis for Gemma: https://docs.google.com/spreadsheets/d/1LA7vHabYONHn00mUdkrlUkM_yufx4TVT/edit?gid=911864116#gid=911864116
- QASA, CLOSEDBOOK analysis for LLaMa: <https://docs.google.com/spreadsheets/d/1NfmsN9R3kW8wAcOt3AIPaEbqHBG6LT5-/edit?gid=1701024737#gid=1701024737>
- ELi5, Vanilla analysis for Gemma: https://docs.google.com/spreadsheets/d/16FXdU1lmibeRX_xYp27Xo3YT_404UFun/edit?gid=1875113499#gid=1875113499
- ELi5, CoT analysis for Gemma: <https://docs.google.com/spreadsheets/d/11QeIQpr78y-P9-WfVdAW2-91B0eIvfsy/edit?gid=1655928492#gid=1655928492>

- ELi5, ToT analysis for Gemma: <https://docs.google.com/spreadsheets/d/1ciVogJEkynxuu2UMcbroBWBWEiTKRQBI/edit?usp=sharing&ouid=104543084022695088057&rtpof=true&sd=true>

A.2 VANILLA w/Chain of Thought Awareness

Instruction: You are given a question and retrieved passages. Each passage block starts with “Passage [rank N]” where N is the citation rank (1, 2, 3, ...) and passage_id=... is the stable QASA passage key from the eval store (same as in qasa_eval.json). Before generating an answer, reason through each passage: note relevance and why. Identify which 1-3 passages are most critical (by rank). Format lines as: “Passage [rank N] (passage_id ...): {RELEVANCE} - {REASON}”.

Instruction: Given your prior analysis (above), write a thorough, accurate, and detailed answer for the question using only the provided search results (some of which might be irrelevant) and cite them properly...

A.3 VANILLA w/Tree-of-Thought Instruction

Instruction: You are given a question and retrieved documents. Simulate three brilliant expert analysts collaboratively answering the question. The experts are: Expert A (Document Analyst: focuses on assessing relevance and reliability of each document), Expert B (Fact Checker: focuses on identifying key facts and potential contradictions across documents), and Expert C (Answer Synthesizer: focuses on how to best combine evidence into a coherent answer).

Round 1 - Each expert assesses document relevance: Expert A: [rates each document High/Medium/Low relevance with reason]
Expert B: [agrees or refines A’s ratings, notes any contradictions or missing info]
Expert C: [identifies which 2-3 documents are most critical for answering and why]

Round 2 - Each expert proposes key claims

for the answer: Expert A: [proposes claims supported by the most relevant documents, with citation numbers]

Expert B: [fact-checks A's claims, adds or removes based on document support, with citation numbers]

Expert C: [refines and synthesizes into a coherent set of claims, with citation numbers]

Final Answer: Based on the expert discussion above, write a single fluent paragraph answer. Use an unbiased and journalistic tone. Always cite for any factual claim using [1][2][3]. Cite at least one document and at most three documents in each sentence. If multiple documents support the sentence, only cite a minimum sufficient subset.

A.4 Vanilla Examples in Analysis

Vanilla Example 1

Question: Is hyperparameter optimization performed independently for the two dataset corpora?

Gold answer: Yes, it does appear that hyperparameter optimization for each dataset is performed independently.

Gemma Model Output: Yes, all model hyperparameters were tuned on the respective validation sets of the two corpora [1].

LLaMA Model Output: Hyperparameter optimization is performed independently for the two dataset corpora, as all model hyperparameters were tuned on the respective validation sets of the two corpora [1]. The hyperparameters considered for tuning include hidden layer sizes, depths, initial learning rates, batch sizes, and dropout, and the best model is selected based on the underlined hyperparameters and the qca setup [1]. The experimental setup and results are provided in more detail in Appendix A and Table 5, respectively [1].

Vanilla Example 2

Question: What is the full form of ILSVRC?

Gold answer: Since there is no information about the form of ILSVRC in this paper, this question cannot be answered and requires external knowledge.

Gemma Model Output: based on the provided

documents, the full form of ILSVRC is not mentioned

LLaMA Model Output: The full form of ILSVRC is not explicitly stated in the provided documents [1][2][3]; however, based on the context, it can be inferred that ILSVRC refers to a competition or dataset related to image classification.

A.5 Vanilla w/CoT Examples in Analysis

Vanilla CoT Example 1

Question: In language models, which method would be better for preventing overfitting from batch normalization and dropout.

Gold answer: According to this work, without dropout, a vanilla LM can run the risk of overfitting, which decreases performance.

Gemma Model Output: Based on the provided research, there is no direct comparison or determination as to whether batch normalization or dropout is a better method for preventing overfitting in language models. The documents do note that using dropout is important for preventing overfitting; specifically, a vanilla language model without dropout runs the risk of overfitting on smaller datasets like TREC-6, which can lead to decreased performance [1]. In standard practices for fine-tuning classifiers, both batch normalization and dropout are used together in additional linear blocks [2]. Furthermore, specific dropout rates are applied across various components of the model such as 0.4 to layers, 0.3 to RNN layers, and 0.5 for weight dropout on the RNN hidden-to-hidden matrix to ensure robust performance across diverse tasks [3].

LLaMA Model Output: To prevent overfitting in language models, particularly when using batch normalization and dropout, the method that would be better is not explicitly compared in the provided passages [1][2][3]. However, dropout is mentioned as a technique to prevent overfitting in a vanilla language model [1], and both batch normalization and dropout are used in the classifier layers following standard practice for computer vision classifiers [2]. The application of dropout to various layers, including layers, RNN layers, input embedding layers, and embedding layers, is described in [3], but this passage does not explicitly compare

batch normalization and dropout. Therefore, based on the information provided, dropout appears to be a method used to prevent overfitting [1][3], while batch normalization is used in conjunction with dropout in the classifier layers [2].

Vanilla CoT Example 2

Question: What point is different in GATs in terms of assigning weight compared to GCN? Gold answer: GATs uses implicit weight assigning while GCN doesn't [3].

Gemma Model Output: Unlike Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs) allow for the implicit assignment of different importances to nodes within the same neighborhood [3]. This ability to assign varying weights to different neighbors enables a significant increase in model capacity [1][3]. Empirical results support the benefit of this approach, as GATs have demonstrated improvements over GCNs on datasets such as Cora and Citeseer by assigning different weights to nodes in a shared neighborhood [1].

LLaMA Model Output: The key difference between Graph Attention Networks (GATs) and Graph Convolutional Networks (GCNs) in terms of assigning weights is that GATs allow for assigning different importances to nodes of the same neighborhood, whereas GCNs do not [3]. This is achieved through the use of attention mechanisms in GATs, which enable the model to weigh the importance of different nodes in the neighborhood [1]. In contrast, GCNs use a fixed weighting scheme, where all nodes in the neighborhood are assigned the same weight [3]. This difference in weighting schemes allows GATs to capture more complex relationships between nodes in the graph, and has been shown to improve performance on various graph-based tasks [1]. The ability of GATs to assign different weights to nodes of the same neighborhood is a significant advantage over GCNs, as it enables the model to learn more nuanced and context-dependent representations of the graph [3]. Overall, the use of attention mechanisms in GATs provides a more flexible and powerful way of assigning weights to nodes in the graph, compared to the

fixed weighting scheme used in GCNs [1][3].",

A.6 ClosedBook Examples in Analysis

CLOSEDBOOK Example 1

Question: Which datasets are used by the paper for supervised learning?

Gold answer: UCF-101 and HMDB-51 datasets are used for supervised learning.

Gemma Model Output: The CelebA and LSUN datasets.

LLaMA Model Output: CIFAR-10 and ImageNet

A.7 ELI5 VANILLA and VANILLA w/CoT Examples in Analysis

ELI5 Example 1

Question: When and why did clapping become a thing?

Gold answer: Please read this entire message

—
Your submission has been removed for the following reason(s):

ELI5 is not for:

Straightforward answers or facts - ELI5 is for requesting an explanation of a concept, not a simple straightforward answer

— If you would like this removal reviewed, please read the [detailed rules]first. If you still feel the removal should be reviewed, please [message the moderators.]"

VANILLA Output: "Based on the provided documents, there is no definitive explanation for when or why clapping became a general practice [1][2][3]. The documents only offer anecdotal observations regarding specific instances of clapping:

* One account describes a classroom setting where students and a teacher engaged in enthusiastic clapping following an explanation [1].

* Another observer notes that clapping occurs upon landing in Kazakhstan, suggesting it may be a regional custom potentially connected to the former Soviet Union [2]. * A third account describes frequent clapping by an audience during a competition, sometimes occurring for no apparent reason [3]."

VANILLA + CoT Output: Based on the provided documents, there is no information explaining the historical origins or the specific

reasons why clapping became a practice. The documents only offer anecdotal observations, such as a mention of clapping potentially being a "regional thing" in Kazakhstan or the former Soviet Union [2], or descriptions of clapping used in classroom and audience settings [1][3].

ELI5 Example 2

Question: What is chirality? I've heard the right-hand, left-hand metaphore but that doesn't make any sense to me. What is actually happening?

Gold answer: If you want to tighten a screw, you turn it clockwise. Now, imagine that you have a similar screw, but it tightens *counter-clockwise* instead of clockwise. We would say that these two screws have different "chiralities".

VANILLA Output: "Chirality refers to the property of ""handness"" [3]. The metaphor of the right and left hands is used because, while they are very similar and related as mirror images, they are not identical and cannot be superimposed on top of each other [3].

In a chemical context, molecules can be chiral if they contain one or more chiral centers [3]. In organic chemistry, a chiral carbon is an asymmetric carbon atom that is bonded to four different groups [1]."

VANILLA + CoT Output: In chemistry, molecules exhibit this property if they contain one or more chiral centers [3]. A common example in organic chemistry is a chiral carbon, which is an asymmetric carbon atom bonded to four different groups [1]. For a molecule to have stereoisomers, it must essentially be chiral [2]."

A.8 ToT Examples in Analysis

ToT QASA Example

Question: What are the motivation behind choosing TREC-COVID for analysis on annotation bias?

Gold answer: This dataset could be made unbiased by manual annotations [2].

ToT Output: "Round 1 - Each expert assesses document relevance: Expert A: Document [1] High relevance - Explains the specific pooling method used in TREC-COVID and intro-

duces the concept of studying bias through the Hole@10 rate. Document [2] High relevance - Explicitly discusses the lexical bias present in datasets and how the analysis of TREC-COVID revealed an unfair disadvantage for non-lexical approaches. Document [3] Medium relevance - Discusses the findings regarding annotation selection bias and the impact on non-lexical approaches, but is slightly more focused on the implications for the BEIR benchmark. Expert B: I agree with Expert A. Document [1] and [2] are essential because they connect the dataset's methodology (pooling) to the specific bias being investigated (lexical/annotation bias). Document [3] reinforces the findings but is more of a summary of the study's conclusions. There are no contradictions; all documents point toward the investigation of how annotation methods affect the fairness of model evaluation. Expert C: Documents [1] and [2] are the most critical. Document [1] provides the technical context of how TREC-COVID was constructed (via pooling), which is necessary to understand the nature of the bias, while Document [2] provides the direct motivation: identifying lexical bias that disadvantages non-lexical retrieval models.

Round 2 - Each expert proposes key claims for the answer: Expert A: Claim 1: TREC-COVID was used to study the impact of annotation selection bias [1]. Claim 2: The dataset utilized a pooling method where the annotation set was constructed from search results of participating systems [1]. Claim 3: Analysis of the dataset revealed a strong lexical bias that can disadvantage non-lexical approaches [2]. Expert B: All claims are accurate. I will add: Claim 4: The researchers found that manually annotating missing relevance judgments for the tested systems led to significant performance improvements for non-lexical approaches, confirming the bias [2]. Claim 5: The study found varying Hole@10 rates across systems, which negatively impacts non-lexical models [3]. Expert C: Synthesized claims: The motivation was to study annotation selection bias [1]. This was investigated using TREC-COVID's pooling-based construction [1]. The goal was to address lexical bias that unfairly disadvantages non-lexical models [2], a finding supported by

the fact that manual annotation of missing judgments improved non-lexical performance [2] and that disparate Hole@10 rates negatively affect these approaches [3].

Final Answer: The motivation for choosing TREC-COVID was to study the impact of annotation selection bias [1]. Because the TREC-COVID annotation set was constructed using a pooling method based on search results from various participating systems, it provided a framework to analyze how such construction affects evaluation [1]. The researchers specifically aimed to investigate lexical bias, which can create an unfair disadvantage for non-lexical retrieval approaches [2]. This investigation was driven by the observation that different systems exhibited significantly different Hole@10 rates, a factor that negatively affects non-lexical models [3]. Ultimately, the analysis confirmed this bias, as manually annotating missing relevance judgments resulted in significant performance improvements for non-lexical approaches [2]."

ToT ELI5 Example

Question: In the middle of yawning, why did the music I was listening to change to a lower pitch?

Gold answer: It changes the pressure of the cochlear fluid, which results in you sensing things differently. This fluid is part of your normal sense of pitch, but that pressure doesn't typically change much ... accept when you yawn.

Provided documents: [1] Sound of Music or Sound of Silence: Why We Find Music Rewarding – Brainy Sundays: we need to look at the musical features that make this art form rewarding. Musical pieces follow a culture-specific structure that allows listeners to create expectations and predictions. For instance, they have well-defined and easily predictable beginnings, middles and ends. Furthermore, Western music follows specific rules: the direction of the next pitch change can be predicted from the current pitch change size; the melody is often arch-shaped, with a rise in the middle and a return at the end; the distribution of tones follows certain statistical rules, e.g., the 5ths occur more often than the 6ths and the 7ths;

and

[2] Music and Brain Blog at University of Toronto: 10/20/13 - 10/27/13: the presence of changes of pitch more accurately than they can perceive the direction of changes in the pitches. Clear and regular harmonic structure in the lower-frequency channels will help cochlear implant users with pitch discrimination. The performance will decrease when there is interference in the high-frequency channels. As a student, I had classmates who were hard of hearing. The teacher would speak through a FM transmitter and they would have an interpreter in the class signing. I recall my classmates received hearing aids as they did a combination of reading our facial expressions

[3] Intonation Definition and Examples in Speech: Everyone has different levels of the pitch in their voice, notes Study.com: "Though some are more prone to a higher pitch and some to a lower pitch, we can all change our timbre depending on who we are talking to and why." Timbre refers to the quality of sound that distinguishes one voice or musical instrument from another or one vowel sound from another: It is determined by the harmonics of the sound. Pitch, then, refers to the musicality of your voice and how you use that musicality or timbre to convey meaning. Chunking—and pausing—meanwhile packages information for the listener.

ToT Output: "Round 1 - Each expert assesses document relevance: Expert A: Document [1] Low relevance - discusses musical structure and predictability but does not mention yawning or physiological changes. Document [2] Low relevance - discusses pitch discrimination and cochlear implants, which is unrelated to the user's experience. Document [3] Low relevance - defines pitch and timbre in speech but does not address the physiological effects of yawning on hearing. Expert B: I agree with Expert A. None of the provided documents contain information regarding the physiological relationship between yawning and pitch perception. There is a complete lack of information regarding how yawning affects the ear, the auditory system, or the perception of sound. Expert C: Since none of the documents address the specific phenomenon of yawning affecting

pitch perception, I cannot identify 2-3 critical documents to answer the question. The provided documents are irrelevant to the query.

Round 2 - Each expert proposes key claims for the answer: Expert A: No claims can be made as the documents do not support the question [1][2][3]. Expert B: I cannot verify any claims regarding yawning and pitch changes because the documents focus on musical structure [1], hearing aids [2], and speech intonation [3]. Expert C: I cannot synthesize an answer because there is no evidence provided to explain why music might sound lower in pitch during a yawn.

Final Answer: Based on the provided documents, there is no information available to explain why music might change to a lower pitch during a yawn. The documents discuss musical structure and predictability [1], pitch discrimination in relation to cochlear implants [2], and the definition of pitch and timbre in speech [3], but they do not address the physiological effects of yawning on auditory perception."